claude-windows / 8d4866bf-8edd-4da9-909e-022ddee12675

Metadata

- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\8d4866bf-8edd-4da9-909e-022ddee12675\subagents\workflows\wf_4e32c8b8-725\agent-a5c2569e341c71ed9.jsonl`
- SHA-256: `87fc196cb3bd9a47c8e6d66d5608f44962803476469d0486922eef84e0c96ffe`
- Source modified: `2026-06-19T17:19:51+00:00`
- Imported at: `2026-07-05T16:48:25+00:00`
- project: `wf_4e32c8b8-725`
- session_id: `8d4866bf-8edd-4da9-909e-022ddee12675`

Transcript

1. user (2026-06-19T17:18:10.919Z)

You are auditing the docs/ folder of the badminton-highlight-indexer repo to decide which docs can move to docs/archives/. Be rigorous and EVIDENCE-BASED ? the owner has run two doc-staleness sweeps and cares deeply about not archiving live work or breaking links.

REPO ARCHIVE POLICY (docs/archives/README.md) ? APPLY STRICTLY:
- Archive ONLY work in a clear TERMINAL state: DONE / COMPLETED / REJECTED / fully SUPERSEDED. "Never archive in-flight or pending work."
- KEEP-LIVE (do NOT archive) anything that is: living/active, in-progress, a not-yet-started but still-intended design/plan (owner-gated future work is LIVE, not terminal), a reference/navigation doc (CODE_MAP, README, HOW_RALLY_DETECTION_WORKS, EVALUATION), a template (PROJECT_DOC_TEMPLATE), or a roadmap/backlog/strategy doc still being tracked.
- Destinations inside archives/: past_projects/ (bounded completed project / tech-debt / starter effort) · research/ (completed or superseded research notes/plans) · decisions/ (ADRs) · checkpoints/ (dated state snapshots) · archives/ top-level (one-off audits/reports/launch records). Naming convention for a completed doc: <NAME>-COMPLETED-<date>.md.
- ALREADY-ARCHIVED COPIES EXIST for these (so the top-level file is likely a pointer-stub or a stale DUPLICATE to reduce, NOT a fresh move): docs/archives/CODE_AUDIT_AND_TEST_HARDENING-COMPLETED-2026-06-14.md, docs/archives/DEFERRED_HARDENING_PLAN-COMPLETED-2026-06-14.md, docs/archives/HARDENING_LOOP_HANDOFF-COMPLETED-2026-06-14.md. If your doc already has an archived copy, check whether the top-level is a stub (leave it) or a full duplicate (flag DEDUPE).
- CONTRADICTION TRAP: a doc may say "COMPLETED" yet still be cited as a LIVE source elsewhere (e.g. CLAUDE.md cites DEFERRED_HARDENING_PLAN.md #16 as the next platform slice). Before recommending ARCHIVE, grep CLAUDE.md, docs/NEXT_STEPS.md, docs/README.md, and other top-level docs for the filename. If it is referenced as a live/next source, verdict = ASK_OWNER (note the contradiction), not ARCHIVE_NOW.

YOUR BATCH (read each from docs/<name>): RALLY_QUALITY_RESEARCH.md, RALLY_DETECTION_GAP_CLOSURE.md, RALLY_DETECTION_QUALITY_REPORT.md, PROMINENT_COURT_DETECTION.md, QUALITY_ITERATIONS.md, R2_EVAL_METRIC_DESIGN.md, MULTI_SIGNAL_FUSION_PLAN.md

For EACH doc:
1. Read the status header + TL;DR/intro + (if present) the §7/§8 progress tracker + definition-of-done. You do NOT need to read every line of long docs ? sample enough to judge lifecycle state honestly.
2. Decide lifecycle and verdict per the strict policy. Remember: owner-gated / not-yet-started DESIGN docs are LIVE future work (KEEP_LIVE), NOT terminal. Only DONE/COMPLETED/REJECTED/fully-superseded is terminal.
3. Check for an already-archived copy (the policy lists three; also Glob docs/archives for the doc's stem). If the top-level is a thin pointer-stub ? KEEP_LIVE (it's intentional). If it's a FULL DUPLICATE of an archived COMPLETED copy ? DEDUPE_STUB (recommend reducing top-level to a stub or removing).
4. Cross-check "cited as live": Grep CLAUDE.md, docs/NEXT_STEPS.md, docs/README.md, and the other top-level docs/ for this filename. If it's cited as a live/next/authoritative source, it

CANNOT be archived even if it says COMPLETED ? ASK_OWNER with the contradiction noted.
5. Inbound references: Grep the whole repo (docs/, backend/, CLAUDE.md, *.py) for the filename to count link-breakage risk; list up to 6 key referrers.

Return the structured object with one entry per doc in your batch. Be decisive: most reference/template/active/design docs are KEEP_LIVE; reserve ARCHIVE_NOW for genuinely terminal single-efforts/reports/completed projects, and ASK_OWNER for the contradictory/ambiguous ones.

2. assistant (2026-06-19T17:18:13.463Z)

I'll audit each doc in my batch. Let me start by reading the archive policy and the key cross-reference files, then read each doc.

3. user (2026-06-19T17:18:14.446Z)

```
1	# archives/
2
3	Umbrella directory for all **historical and completed** material.
4
5	## Live vs Historical
6	- **Live docs** (for ramp-up, constant updates, and current work) live at the top level of `docs/`.
7	- Everything here is no longer the primary source of truth. It is kept for context, decision history, and regression detection.
8
9	## Classification inside archives/
10	- `past_projects/` ? Completed small tech debt, starter projects, or low-regret initiatives (e.g. the original low-regret moves + scaffolding work). These are "past" once they reach a terminal state.
11	- `decisions/` ? Historical ADRs and major architectural decisions.
12	- `research/` ? Completed or superseded research notes and plans.
13	- `checkpoints/` ? Snapshots of major project states (especially useful for detecting regressions in quality, priorities, or baselines over time as we do quality iterations).
14	- Other one-off audits, etc.
15
16	## Archival Rule
17	Only move something into `archives/` (or its subdirectories) when the work is in a clear **terminal state**:
18	- DONE
19	- COMPLETED
20	- REJECTED
21
22	Never archive in-flight or pending work.
23
24	When archiving a project/phase:
25	1. Ensure the main document has an explicit `**STATUS: COMPLETED**` (or DONE/REJECTED) header at the top.
26	2. Link back to the current living documents in `../` (e.g. `COMMERCIALIZATION.md`, `BACKLOG.md`).
27	3. For regression detection value, consider adding a dated entry under `checkpoints/`.
28
29	See `past_projects/README.md` for the specific guidance on that category.
30
```

4. user (2026-06-19T17:18:15.475Z)

```
1	# Next steps (live) ? owned-model + multisport
2
3	**Updated:** 2026-06-13 (**rally-detection quality track**: fusion + experiment-harness design docs merged,
4	owner-gated ? see the dedicated section below, the desktop-agent pick-up). Prior: 2026-06-10 (audit?M0?Gemini-battery session; later same day the **UI-alpha track opened**:
5	ADR-009 ratified ? KhelSutra Guru / khelsutra.guru, private `khelsutra` repo, local-first delivery ? see
```

`docs/UI_ALPHA_EXECUTION_PLAN.md`; **PR-1 async jobs (#16) MERGED 2026-06-11, PR #87**; **PR-2 upload/download MERGED 2026-06-11, PR #99** ? live-E2E-verified (3-part chunked upload MD5 byte-identity + compile?browser-download); next: PR-3 identity. Same day: the **8?9 quality sweep #90?#98 landed** ? ruff/mypy/ coverage-floor CI lanes, SQLite conn-close fix, config unknown-key warning, hybrid-twin collapse into `trajectory_hybrid.py`, WSL tilde+abspath live-path fixes; suite 388 green; LOVO 0.626 re-verified exact).
Authoritative roadmap: `docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md` (+ ADR-008 for topology). **Latest blow-by-blow:** memory `current-state.md` / `session-handoff.md`. This file = the prioritized "what to do next," refreshed each session.

? Next-session / GPU-box pickup (2026-06-15)
M0 in-container quick wins are COMPLETE: ship-gate provisional blocker (**#170**), train/eval split guard (**#171**), the **doc-consistency sweep** (**#174** ? corrected 16 drifted docs vs the code), and the **P3 person-feature fusion litmus machinery is already present + suite-green** (`backend/eval/fusion_compare.py` + `ablation.py`: person-driven ?F1 / bootstrap CI / paired sign test; the real-golden `skipif` litmus waits only on GPU-extracted data). ? CI is red repo-wide due to a **GitHub Actions minutes/billing outage** (account-level, not code) ? work is local-gated + `--admin`-merged until Actions is restored (Settings ? Billing ? Actions minutes / spending limit).

Immediate next step ? the real P3 LOVO measurement (this box has a GPU + `torch+cu124` + `cv2`; videos and trajectory CSVs are present, but the GPU-extracted golden features + `.rallies.csv` labels are not yet here):
1. **[owner / GPU box]** `python -m backend.eval.fusion_golden --manifest output/human_lovo_manifest.json --out-dir "%USERPROFILE%\OneDrive\rally-annotator\golden-data\features\2026-06-15-n6"` ? produces `*_golden_features.json` for the 6 golden videos. (GPU + WSL + videos + trajectory CSVs + `.rallies.csv` labels are owner-side only.)
2. Artifacts land in the **shared cloud folder** `C:\Users\avidu\OneDrive\rally-annotator\golden-data\` (OneDrive auto-syncs both ways). Convention + workflow: **[`GOLDEN_DATA_SHARING.md`](GOLDEN_DATA_SHARING.md)**; tracked in **#175**.
3. **[CPU dev/agent session]** `python -m backend.eval.fusion_compare --features-dir <shared>\features\2026-06-15-n6 --record` + `python -m backend.eval.ablation --features-dir <shared>\... --granularity group` ? real person-feature ?F1; copy the JSONs into `output/` to also pass the `skipif` litmus tests (`tests/test_ablation.py::test_litmus_reproduces_gate_a`).

Also queued (owner-gated / pending input):
- Ship-gate target calibration ? **#172** (owner decision; needs corpus growth).
- Tests reorganization ? blocked on the owner's grouping metric (provide it ? an agent will do it).
- Restore CI ? the Actions outage is account-level (billing/minutes), not a repo/config bug.

Leaving (bulky / M2 ? no start without owner go): ActionFormer (M0.4), event-index DB migration (M0.1), Gemini-silver purge (M0.3), cost-to-Gen-2 benchmark + ByteTrack deprecation, multisport, repo rename, PMF execution.

Where we are (one paragraph)
ADR-008 ratified ("repo split driven by productionization, not isolation"): training lives in-repo (`training/`) behind a CI-enforced import wall; the featurizer is single-sourced (`backend/features/rally_seq.py`, train==serve); the **Gen-0 harness ships** (`python -m training.gen0.harness --manifest output/human_lovo_manifest.json --floor eval_baselines/heuristic_lovo_n6.json --version <semver>`) and produced artifact v0.1.0 (LOVO **0.626**, floor passed, sign test 5/6 wins p=0.109 ? honestly not significant, ship=False). The **n=6 owned corpus is in the collector** (262 rallies, Proprietary, commercial export = owned data only after the ShuttleSet reclassification). The **Gemini

battery is done** (see table) ? the moat is quantified.

The quality table that now drives strategy (IoU 0.5 vs human golden labels)

Path	Mahadevpura (clean-ish)	Kushagra (multi-court)
Heuristic (velocity windowing)	0.657	0.157
Gen-0 owned head (LOVO, n=6)	0.744	0.561
Two-stage WASB+Gemini refine	0.83	?
Gemini wrapper (`gemini-flash-latest`)	**0.93**	**0.66**

Reading: clean footage is commoditized (serve it with Gemini for cents). **Multi-court drops the commodity wrapper to 0.66 ? that is the owned model's ship target** (its FPs are background-game pickups + dead-time merges, the exact contrast-metric confound). Gen-0 is 0.10 behind with only n=6 ? a corpus-growth-sized gap, not an architecture-sized one. Two-stage refine = the cost-efficient serving path for LONG tapes (uploads candidate clips only); wrapper 503-flakiness (observed all session) makes the provider-fallback registry load-bearing.
Baselines tracked in `eval_baselines/` (heuristic_lovo_n6.json, gemini_wrapper_2026-06-10.json).

Rally-detection quality track (design merged 2026-06-13, owner-gated; implementation in progress)

This is the track a desktop/local agent is slated to pick up. Design docs landed (PR #112 + session). All implementation is **owner-gated** (per explicit in the plan); each step = ONE PR off `master`.

Hardening loop (T1-T5 positive tests + audit + finale) COMPLETE (2026-06-14, PRs #156/#157/#160/#161/#163 + #165 + final records). See archived `docs/archives/CODE_AUDIT_AND_TEST_HARDENING-COMPLETED-2026-06-14.md` (Living Log with every pre/post/review) and `docs/archives/HARDENING_LOOP_HANDOFF-COMPLETED-2026-06-14.md` (full rules + STATE). Top-level now pointer stubs per #165 review. All 5 items + verification + archive done; clean slate.

- **Docs:** `docs/MULTI_SIGNAL_FUSION_PLAN.md` (shuttle + person + audio, one fusion layer) · `docs/EXPERIMENT_HARNESS.md` (A/B + shadow + promotion gate, zero-regression) · `docs/ROORA_LABELING_GUIDELINES.md` (v8) · `docs/HOW_RALLY_DETECTION_WORKS.md` (2-layer mental model).
- **Key finding (no re-investigation needed):** the owner's "held-shuttle counted as a rally" bug is **NOT a missing capability** ? `backend/pipeline/detectors/rally_gate.py` already implements the net-crossing **"Gate A"** (`apply_gate(..., min_crossings=2)`) that kills service-feed/held-shuttle windows, but it is **only called from eval drivers** (and there was no `min_crossings` knob in `IndexingConfig`). **Wiring Gate A into the live segmenter (now via fusion_hybrid) was the single cheapest, highest-confidence quality win.** (P2 FusionSegmenter specifics: registry `fusion_hybrid`, **default-OFF**, seeds from substrate, persists `net_crossings` + optional person features, optional served Gate A/B via config knobs; adversarially reviewed; suite green. See `tests/test_served_gate_a_regression.py` for the honest finding that on production windowing Gate A is neutral/negative in some cases, hence kept opt-in with knob for safety, and the leaderboard revert note.)
- **Current status (2026-06-14):** P1 (person cue extraction to reusable `PersonDetector`) + P2 (FusionSegmenter + Gate A signal + served handoff gating) + experiment harness P1 DONE (adversarial review + NO-REGRESSION; suite green). **P3 next (this PR + follow-ups):** learned fusion (person features ? harness `compare` LOVO lift on the 6 golden videos; eager-person-compute optimisation); then P4 audio via hit_detector.
- **Pick-up order (each = ONE PR off `master`, owner approval first, flag-gated default-OFF, promote only on measured LOVO):**
  1. (P2 first-step, largely done via fusion_hybrid) Wire Gate A into production served path + config knob (min_crossings); apply in hybrids at runtime. (Now available opt-in; served neutral on production windowing per litmus ? kept opt-in with knob for safety.)
  2. P3 first step (this work item): harness `compare` LOVO litmus for person-on variant (players_in_court + two_sided + colocation etc. from `fusion_features`) on the 6 golden; add the eager compute optimisation note/landing. Pure + harness (testable here).

3. P3/P4 continuation: audio cue integration + full learned promotion if lift proven.
- **Discipline (from EXPERIMENT_HARNESS.md):** every new cue **flag-gated, default OFF**; promote to default **only on measured LOVO lift** through `run_regression.py` (exit 0/1/3); pinned `CONTROL` ? rollback = config flip. Harness re-uses ~80% existing code.
- ? Gate A, person cue logic, harness, and pure fusion paths are testable here; the heavier real-video / golden LOVO / ablation confirmation waits on the GPU-extracted golden features + `.rallies.csv` labels, which are not yet present in this checkout.
- **External due-diligence review inputs (2026-06-17) ? PICK UP AFTER current ideas are exhausted; full details + citations in [`RALLY_QUALITY_RESEARCH.md`](RALLY_QUALITY_RESEARCH.md) §5.6.** An owner-commissioned 31-reference review of the quality report *validated* the methodology and surfaced these **future** levers (research inputs, not active work, not new claims; verify citations before any submission):
  1. **"First-mile" paper framing (highest-value):** stroke-classifiers (BST/DSTA) + the big datasets (ShuttleSet ~36k strokes, Fine-Badminton ~27k actions) all **assume a pre-trimmed rally clip already exists** ? we solve the ignored prerequisite. Pitch the paper as *"the missing untrimmed-video preprocessing pipeline that lets BST/MonoTrack run on raw footage."*
  2. **Ground R2 in Action-Spotting / Precise-Event-Spotting** (SoccerNet/TenniSet/T-DEED) ? turns R2 into a defensible publishable contribution.
  3. **Add domain baselines for any submission:** MonoTrack + BST on ShuttleSet / Fine-Badminton (we have only heuristic + Gemini today).
  4. **Prefer TemporalMaxer over ActionFormer** for the M0.4 scorer swap (parameter-free max-pool, ~2.8× fewer GMACs, small-N robust) ? see "M0.4 next increment" below.
  5. **Court polygons ? pose ? the rally-END lever:** masking on-court players can resolve *occluded end-of-rally states* (the #1 remaining gap vs Gemini), not just spectator noise.
  6. **Name the eval?serve lesson "pipeline distribution skew"** ? a citable paper "lesson."

Discipline reminder: owner approval before starting any owner-gated item; one PR per step; babysit the PR (poll/fix/reply) until merge observed, *then* advance.

Prioritized next steps
1. **[owner, in progress] GROW THE GOLDEN CORPUS** ? the single highest-leverage move; every lever below feeds on it.
   Priority order for new videos: **(a) multi-court badminton** (the target distribution AND where the ship target lives), **(b) ?3 matches per new sport ? table tennis first** (WASB transfers at F1 0.909; pickleball labels are corpus-gold but not trainable until a clean MIT/Apache ball detector exists), (c) capture variety per `VIDEO_RECORDING_GUIDELINES.md`; **adult sessions only for the training pool** (consent guardrail). Per video: label ? `curate import-local --normalize --license Proprietary --fps <true fps>` ? re-run the Gen-0 harness. The paired sign test needs n: at n=10 with 9 wins, p?0.011 ? significance is reachable this month.
2. **[owner decision] Set the ship-gate target** ? now informed: floor = heuristic 0.480 (necessary), **multi-court reference = wrapper 0.66** (the number that makes the owned model worth serving), clean-footage ceiling = 0.93 (not the target ? Gemini serves that tier). Suggested shape: "LOVO F1@tIoU0.5 ? 0.70 on multi-court folds with paired-sign p<0.05 vs heuristic" ? owner to ratify/adjust.
3. **M0.4 next increment ($0/local):** swap the Gen-0 LogReg for a lightweight temporal head (MIT, minutes on the 2070S ? NOT the heavyweight Gen-2 stack) inside the existing harness; per-video threshold calibration (the LOVO-vs-oracle gap is visible per fold in the harness output). **Prefer TemporalMaxer** (parameter-free max-pool, ~2.8× fewer GMACs, small-N robust) over ActionFormer/ASFormer per the 2026-06-17 external review (`RALLY_QUALITY_RESEARCH.md` §5.6) ? better fit for the cheap/local/fast mandate.
4. **Multisport thread:** ingest Roora pickleball/TT CSVs (`import-local --normalize`); merged-prototype LOVO across badminton+TT once TT has ?3 matches; identify a clean pickleball ball detector. ? TT-

on-WASB serving stays gated
99 by C3 (provenance multiplies per sport).
100 5. **PMF validation (cheaper than any engineering month):** C13 academy interviews (+ the two added questions:
101 "multi-court footage nobody watches?" / "pay per-match to index it?"); camera pilot at the warmest 2-3 academies
102 (consent at capture; demo served via the Gemini path or human-polish ? the reel pattern exists:
103 `output/MahadevpuraSingles_highlights.mp4`). **Field kit COMPLETE (2026-06-12, owner ratified the
104 physical-visit approach; reframed product-feedback-first per owner spec):**
105 `docs/PRODUCT_FIELD_ASSOCIATE_FLYER.md` (the recruiting flyer ? owner fills ?/contact placeholders and recruits) +
106 `docs/FIELD_VISIT_PLAYBOOK.md` (the hired associate's visit playbook: coach + student script, demo-after-Q6
107 rule, ground rules, Phase-2 record-and-show-back stretch goal) + `docs/FIELD_VISIT_ANSWER_SHEET.md` (per-visit
108 capture form incl. demo-reaction + student blocks) + `docs/PMF_FIELD_SIGNALS.md` (G/A/R anchors + **pre-registered
109 decision rules** ? PROCEED / PIVOT A-C / FALSIFIED thresholds fixed before the first visit).
110 6. **Housekeeping:** ~~rotate the Gemini API key~~ **DONE 2026-06-10** (owner-verified working; confirm old-key
111 deletion in AI Studio); **rename repo off "badminton"** (owner action,
112 in BACKLOG); collector=SoR confirmation; off-peak niceties: none pending ? the battery completed.
113
114 ## Pointers / artifacts
115 - Roadmap: `docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md` · Topology: ADR-008 (`docs/archives/decisions/DECISIONS.md`) ·
116 Study: `docs/archives/research/OWNED_MODEL_TRAINING_STUDY.md` · Quality history:
117 `docs/archives/quality-iterations/2026-06-multivideo-lovo/`.
118 - Harness: `training/gen0/harness.py` (one command ? train ? LOVO ? gate ? artifact bundle; weights gitignored,
119 manifest reviewable). Featurizer: `backend/features/rally_seq.py` (FEATURE_SCHEMA v1).
120 - Corpus: collector DB (system-of-record, 6 matches/262 rallies) · eval manifest `output/human_lovo_manifest.json`
121 (gitignored) · durable trajectories + Roora files: `C:\Users\avidu\Projects\Annotation Setup\Collect\Trajectories\` ·
122 labels: `...\Collect\Golden Labelled\`.
123 - Gemini ops learnings: serial uploads (`indexing.ai_max_workers=1`) + chunk re-encode
124 (`indexing.ai_chunk_scale_width=1280`) + 6-attempt exponential generate retry ? a cold/prepaid project burst-throttles
125 with a MISLEADING "credits depleted" 429; ramp gradually.
126 - ? No owned-model implementation beyond greenlit M0.x until owner greenlights the next milestone. Committed next
127 PLATFORM slice = async job model (single-tenant).
128

5. user (2026-06-19T17:18:15.509Z)

1 - ROORA_LABELING_GUIDELINES.md ? Latest (v8) vendor labeling guidelines mirrored from the collector's `docs/annotation/`, with the fusion-driven court-calibration / singles-doubles additions marked `? ADDED (fusion)`.
2 - RALLY_QUALITY_RESEARCH.md ? **? Foundation doc** for closing the local-vs-Gemini rally-detection gap: the measured gap + the code-grounded over-merge diagnosis ("shuttle moving ? rally"), an inventory of the eval/experiment harness + its gaps, the strengthened evaluation framework (segmental Edit/F1@k that make over-segmentation first-class, golden-corpus + active learning, Gemini-agreement as a *shadow* signal), an **adversarially-verified, cited external-research synthesis** (TAL/TAS boundary heads, racket-sports rally-state cues, small-data, eval rigor), a **prioritized cheap?durable roadmap with falsifiable success criteria**, and (§7) a **tracked project plan** ? work-item checklist, sequencing, the single quality **number** attributed to the effort, and a **definition of done** (project completes when all tiers land + the number is recorded). The plan that the QUALITY_ITERATIONS.md log executes against; ties together

EXPERIMENT_HARNESS / MULTI_SIGNAL_FUSION_PLAN / ANALYZERS_AND_RECONCILERS / OWNED_MODEL_IMPLEMENTATION_PLAN.
3 - RALLY_DETECTION_GAP_CLOSURE.md ? **Tracked execution doc (2026-06-18)** for closing the remaining rally-detection accuracy gap **per video, before launch**, driven by the growing golden corpus. First ground-truth clip (GX010141) reframed it: boundaries are tight (R2/R3/R4 holds) ? the gap is **detection**: (G1) recall (local & Gemini both miss ~half on hard clips, and *different* rallies ? complementary), (G2) precision (local over-fires), (G3) **rally-END over-extension on the quick-return-after-point** (shuttle still moving ? needs a *player-state* signal). Levers (owner-gated): player-kinematics + shuttle-in-hand / sustained-invisibility end-rule with reverse-timeline backtrack, `rally_gate` Gate A wiring, complementary fusion. **Peer** of `RALLY_QUALITY_RESEARCH.md` (execution tracker, not research); §7 is the source of truth.
4 - PROMINENT_COURT_DETECTION.md ? **Design + experiment (2026-06-19, default-OFF, owner-gated flip)** for the **multicourt G2 over-fire**: identify the **prominent (foreground) court** (the one covering the majority of the frame) and keep only rallies whose shuttle lives inside it ? re-defining the shuttle signal `(x,y,visible)` ? `(x,y,visible-AND-inside-prominent-court)` *before* windowing, so a background/adjacent-court shuttle never forms a rally. 2D floor polygon (MVP) ? **pseudo-3D play-volume** (refinement, so high clears aren't clipped). Born from the GX010142 rally-#1 background-court FP; reuses `fusion_features.point_in_polygon` + `FusionConfig.court_polygon`. §8 progress tracker is the source of truth.
5 - RALLY_DETECTION_QUALITY_REPORT.md ? **Technical-paper checkpoint (2026-06-17)** of rally-detection quality: the honest measured state (over-segmentation milestone tolerance-F1 0.091?0.323 at n=11, p=0.002; merge-rate 0.694?0.150; first real-footage ground-truth at-par with START at Gemini parity), the task-faithful R2 metric, a detailed Gemini comparison, next steps + expected wins, areas yet to be explored, and a publishability / path-to-submission verdict. Renders to a polished PDF; the externally-shareable "toward a paper" summary of `RALLY_QUALITY_RESEARCH.md` + `QUALITY_ITERATIONS.md`.
6 - EVALUATION.md ? How we measure and improve quality.
7 - EXPERIMENT_HARNESS.md ? Design (P1 SHIPPED) for A/B + shadow + SxS experimentation on rally-detection variants with **zero production regression** ? experiments are decoupled from deploy (new cues default OFF, promote only on measured LOVO lift, rollback = flip a flag). Composes the existing `backend/eval/` machinery (`calibration`, `regression`, `metrics`); implemented in `backend/eval/experiment.py`.
8 - QUALITY_ITERATIONS.md ? **Living log** of rally-detection quality iterations (hypothesis ? what shipped ? measured/expected effect ? lesson). The reverse-chronological record; terminal-state snapshots get archived under `archives/quality-iterations/`.
9 - ANALYZERS_AND_RECONCILERS.md ? Design (owner-gated, NOT started): a **signal-fusion framework** for rally detection ? producers emit confidence-scored **signals** at three temporal **scopes** (frame detectors / window analyzers / session analyzers), a learned **scorer** weights them, and an auditable **report-first reconciliation** pass lints the decision against the evidence. The spine: **no special-casing in the decision path** (signals?scorer, never `if/else`). Worked examples: a service-fault analyzer, a warm-up/practice span detector, and consuming a per-frame shuttle-state detector. A peer to `EXPERIMENT_HARNESS.md` and `MULTI_SIGNAL_FUSION_PLAN.md` (ADR-007 modular rally layer).
10 - PERSONALIZATION_PLAN.md ? Design (owner-adopted **hybrid**; Phase-0 landing) for the **per-user personalization feature on a generalization-first baseline**: a thin per-user *tuning* layer gated by the honest small-N statistics (a **min-N promotion floor** + a **structural-guard exemption**, never a service gate) + a **contribution flywheel** (free-tier videos pool into the owner-trained baseline; *"a tool that gets better the more you help it"*). Records the **locked owner decisions** (identity, storage, `min_n` semantics, Gate-A default, free-tier pooling). Built so far: per-user promotion gate (#141) + held-out-USER learning curve (#142).
11
12 - GOLDEN_REGRESSION_FIXTURES.md ? **Design (owner-gated, NOT started):** video-free, non-attributable regression fixtures so a clean clone runs the rally-seg suite with **no video/GPU/DB**. Two layers ? the post-detector freeze + **dual gates** (characterization + quality-floor) reusing the `backend/eval/` stack, and the **SEALED-FIXTURES** privacy filter (de-attribution-at-source, tiered public/key-gated, shuttle-only public tier). Carries a cited **IN/US/EU legal memo** (derived-data IP/privacy), a threat model, and a **§7 progress tracker + definition-of-done**. Extends GOLDEN_DATA_SHARING.md.
13
14 ## Product / platform plans

- DESKTOP_APP_PLAN.md ? **Tracked scoping doc (2026-06-18, owner-gated, `[design]` ? NOTHING built):** package the local pipeline as a **cross-platform desktop app** for non-technical players ? plug in the camera's USB/SD card, click once, get back **only the highlight reels**, fully on-device (no upload, no original-video bloat). The centerpiece is the **de-WSL native-compute port** (run WASB natively per-OS: Linux/Windows CUDA · macOS MPS/CPU ? drop WSL2), and its **make-or-break unknown**: is CPU acceptable on a typical *no-GPU* enthusiast laptop (WASB is ~3.4× real-time on a laptop 2070S)? §7 tracker = **P0 compute-feasibility spike ? P1 native detector ? P2 app shell ? P3 packaging/installers**. Realizes the **L0 fully-local** tier of VIDEO_LOCALITY_MODEL.md; is the **`LocalDesktop` ComputeBackend** of PLATFORM_ARCHITECTURE.md §3.2; the smaller/export-clean owned model (OWNED_MODEL_IMPLEMENTATION_PLAN.md) is one answer to the compute crux.

Practical / ongoing
- TRACKNET_WSL_SETUP.md
- Golden set docs (`GOLDEN_SET_*`)
- Evaluation harness and calibration drivers (in `backend/eval/`)
- GOLDEN_DATA_SHARING.md ? How the GPU box and CPU dev/agent sessions share golden artifacts (trajectory CSVs, `*_golden_features.json`) via a OneDrive folder; the GPU-extract ? CPU-litmus split. See **#175**.

Conventions & templates
- PROJECT_DOC_TEMPLATE.md ? Reusable skeleton for a **tracked project doc**: status header · decisions-locked · progress tracker (?/?/?) · definition-of-done · archived on completion. Standardizes the emergent pattern in `RALLY_QUALITY_RESEARCH.md` §7 and `GOLDEN_REGRESSION_FIXTURES.md`. Use for any substantial design/project work.

Historical documents
Completed plans, old research, ADRs, and checkpoints live in [`archives/`](archives/).

- CODE_AUDIT_AND_TEST_HARDENING.md ? **COMPLETED** (2026-06-14; T1?T5 + full loop + verification per #162). Living record of the June 2026 plumbing / contract / architectural audit, 626-test infrastructure, positive test plans for the 5 identified issues, multi-layer regression protocol (including ablation/experiment layer), and pre/post-change verification checklist. See `docs/archives/CODE_AUDIT_AND_TEST_HARDENING-COMPLETED-2026-06-14.md` and `docs/archives/checkpoints/2026-06-14-hardening-loop-complete.md` for the final archived state. (Top-level is now pointer stub per review; full content in archives/.)

See `archives/README.md` for the umbrella policy and `archives/past_projects/README.md` for classification of small tech debt / starter projects.

6. assistant (2026-06-19T17:18:17.597Z)

Now let me read each doc in my batch. Let me start with the first few.

7. user (2026-06-19T17:18:18.568Z)

Rally-Detection Quality: Research, Eval Framework & Project Plan

Status: Foundation doc (2026-06-16), run as a **tracked project** (§7) ? *"Bridge the local?Gemini rally-detection gap,"* **completed when all committed tiers land and a single quality number is attributed (§7.1).** The empirical + research backbone for closing the gap between the **local, no-cloud rally segmenter** and the **Gemini** path. Iterations land as their own measured PRs, logged in [`QUALITY_ITERATIONS.md`](QUALITY_ITERATIONS.md); this doc is the plan they execute against.

How to read this: §1 frames the gap with measured numbers. §2 is the precise, code-grounded diagnosis (read before proposing fixes). §3 inventories the eval/experiment harness we

already have. §4 specifies how we strengthen that harness (the "really strong evaluation framework").
§5 is the cited external-research synthesis (4 dimensions). §6 is the prioritized roadmap with falsifiable success criteria. **§7 is the project plan ? work-item tracker, sequencing, the single number we attribute to the effort, and the definition of done.** §8 recaps the non-negotiable constraints + cross-links.

> **This doc does NOT duplicate prior design.** It builds on, and cross-links:
> [`HOW_RALLY_DETECTION_WORKS.md`](HOW_RALLY_DETECTION_WORKS.md) (the 2-layer mental model),
> [`EXPERIMENT_HARNESS.md`](EXPERIMENT_HARNESS.md) (the A/B + LOVO harness, P1 shipped),
> [`MULTI_SIGNAL_FUSION_PLAN.md`](MULTI_SIGNAL_FUSION_PLAN.md) (shuttle+person+audio fusion),
> [`ANALYZERS_AND_RECONCILERS.md`](ANALYZERS_AND_RECONCILERS.md) (the signal-fusion framework +
> §13 literature review ? **already** establishes that our skeleton is standard TAL/TAS +
> late-fusion; we *cite, don't re-derive*), [`ABLATION_LAYER.md`](ABLATION_LAYER.md),
> [`EVALUATION.md`](EVALUATION.md), [`OWNED_MODEL_IMPLEMENTATION_PLAN.md`](OWNED_MODEL_IMPLEMENTATION_PLAN.md)
> (TAL-head-first ? VLM), and [`GOLDEN_SET_VENDORING.md`](GOLDEN_SET_VENDORING.md) (corpus growth).

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1. The gap (measured)

Two paths produce rally segments today. **Gemini** (cloud VLM, watches the video semantically) is strong; the **local** path (WASB shuttle trajectory ? velocity windowing, no cloud) lags badly. Measured F1 vs golden labels ([`QUALITY_ITERATIONS.md`](QUALITY_ITERATIONS.md)):

Path	Mahadevpura (clean, single court)	Kushagra (multi-court)
Heuristic velocity windowing (local today)	0.657	**0.157**
Gen-0 owned head (LogReg on 5 trajectory features, LOVO n=6)	0.744	0.561
Two-stage WASB + Gemini refine	0.83	?
Gemini wrapper (gemini-flash)	**0.93**	**0.66**

The local heuristic is **footage-fragile**: fine on a clean single court, collapses on multi-court / busy venues. The owned Gen-0 head already closes most of the multi-court gap (0.561 vs 0.66) ? the residual is **corpus-sized, not architecture-sized**.

1.1 The 2026-06-16 local-vs-Gemini divergence run (no golden labels)

A fully-local run (`wasb_hybrid --skip-ai-handoff`) vs Gemini on 3 fresh matches (Kushagra/Yash June-12; report: `?/June 12 Local-vs-Gemini/_LOCAL_vs_GEMINI_REPORT.md`, generator `scratch/compare_local_vs_gemini.py`). **Gemini is a strong reference here, NOT ground truth** (these clips are unlabeled) ? so this is an agreement/divergence study:

	Gemini	Local (WASB no-AI)
rallies / windows (3 videos)	59	12
total "play"	6:28	**17:41**
mean window duration	**6.1 s**	**65 s** (10.6×)
true misses (gemini-only)	?	**0**
merge groups (one local window ? ?2 Gemini rallies)	?	7

8. user (2026-06-19T17:18:19.332Z)

Rally Detection ? Closing the Accuracy Gap (golden-eval-driven, per-video)

> **Status:** `DRAFT` ? implementation **owner-gated** · **Owner:** Avi · **Created:** 2026-06-18 · **Last updated:** 2026-06-18
> **Lifecycle:** `DRAFT ? IN PROGRESS ? DONE ? archived` (move to `docs/archives/` when DONE)
> **Tracking anchors:** §7 progress tracker is the source of truth; pointer in memory `current-state`; add to `docs/README.md` + `docs/NEXT_STEPS.md` when work starts.
> **Relation to existing docs:** **peer-of** [`RALLY_QUALITY_RESEARCH.md`](RALLY_QUALITY_RESEARCH.md) (that is the research + eval-framework doc; this is the *execution tracker* for the remaining accuracy gap). **Builds on** [`MULTI_SIGNAL_FUSION_PLAN.md`](MULTI_SIGNAL_FUSION_PLAN.md) (signal mechanics), [`QUALITY_ITERATIONS.md`](QUALITY_ITERATIONS.md) (the R-log / empirical backbone), [`GOLDEN_VIDEOS.md`](GOLDEN_VIDEOS.md) (the corpus). Does **not** duplicate them ? it points.
> **Honesty note:** each claim tagged `[verified]` (checked vs code/measured) / `[design]` (proposed). The headline numbers below are real measurements, but `n` is small ? read §6.

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0. TL;DR
We are using the **growing golden corpus** (human rally labels) to find and close the rally-detection accuracy gap **per video, before launch** ? and the first ground-truth clip already reshaped the picture. The boundary work (R2/R3/R4) **holds** ? when local fires on a real rally, its edges are tight (|?| < 1 s). **The remaining gap is DETECTION, not boundaries:** (G1) **recall** ? both local and Gemini miss ~half the rallies on hard clips; (G2) **precision** ? local over-fires (false positives), especially on continuous drilling; (G3) **rally-END over-extension** ? when a player *quickly knocks the shuttle back after a point*, shuttle-velocity alone can't tell "rally" from "casual reset," so the window runs long / merges. n=2 sharpens the priority: **LOCAL = high recall / low precision, GEMINI = high precision / low recall** (opposite, complementary ? on the normal clip local's recall *beats* Gemini's). So **the #1 lever is precision** (cut local's false positives via `rally_gate` Gate-A + colocation; the player-state rally-END rule feeds this), with **signal fusion** (local's recall + Gemini-style precision) as the durable win. **Caveat:** two golden clips so far (n=15 corpus, GX010141+GX010137) ? directional, not conclusive.

1. Problem & goal
The detector is good enough to ship a highlights product, but the golden eval shows clip-dependent accuracy that we want to **close consistently on every video, not just on average** ? catching the failure modes in the quality loop instead of after launch.

- **Concrete target:** on every corpus video, local is **at-par-or-better than Gemini vs the golden labels**, with **no per-video regression** on either eval set (the dual-eval-set discipline, [[eval-dual-set-regression]]).
- **"Good" =** the three gaps below closed enough that a milestone flip (default-ON) clears the over-seg guard AND improves per-video F1 broadly, recorded in a Launch Report ([[launch-report-convention]]).
- **Read first to get current:** [`RALLY_QUALITY_RESEARCH.md`](RALLY_QUALITY_RESEARCH.md) §1?2 (the gap + diagnosis), [`MULTI_SIGNAL_FUSION_PLAN.md`](MULTI_SIGNAL_FUSION_PLAN.md) §4 (in-flight vs in-hand), [`HOW_RALLY_DETECTION_WORKS.md`](HOW_RALLY_DETECTION_WORKS.md) (the 2-layer model).

2. Decisions locked
#	Decision	Source / date	Implication
D1	The gap is **DETECTION (recall + precision + rally-END), not boundary precision**	golden GX010141, 2026-06-18 `[verified]`	Don't build a generic boundary head; spend on rally-state signals.
D2	Measure **per-video AND aggregate; no regression on ANY video**	[[eval-dual-set-regression]] / [[decision-rigor-norm]]	A win must broadly help, not trade one clip for another.
D3	**Golden labels = ground truth; Gemini = reference, not truth**	n=15 corpus	Gemini under-detects too ? agreement ? accuracy. Report both, gate on golden.

| D4 | New signals land **default-OFF + flag-gated**, promoted only on measured lift | [`EXPERIMENT_HARNESS.md`](EXPERIMENT_HARNESS.md) / [[weak-signals-retained]] | Zero-regression rollout; retained even on a small-n null. |
| D5 | **Rally-END needs a PLAYER-state signal** ? shuttle velocity can't separate a competitive rally from a casual post-point return | owner insight + R4 limitation, 2026-06-18 `[design]` | Add person kinematics / shuttle-in-hand as the rally-end discriminator. |

3. Foundation ? the measured gaps (golden ground truth)
Ground-truth-scored Boxhill clips (more land as the owner labels them). Milestone windowing = cap6 + R4 ([`QUALITY_ITERATIONS.md`](QUALITY_ITERATIONS.md) R3/R4). Local & Gemini vs truth, n=2 so far: `[verified]`

clip (golden)	system	found	tIoU@0.5 P/R/F1	tolF1	hits	false+	missed	|?start|	|?end|
GX010141 (29, hard/drilling)	LOCAL	22	0.59/0.45/**0.51**	0.43	13	11	16	0.78	0.68
	GEMINI	16	0.94/0.52/**0.67**	0.53	15	4	14	0.31	0.46
GX010137 (34, normal match)	LOCAL	39	0.77/**0.88**/**0.82**	0.68	30	14	9	0.39	0.44
	GEMINI	27	1.00/0.79/**0.89**	0.89	27	0	7	0.50	0.52

- **Corpus aggregate (n=13, milestone):** tIoU F1@0.5 **0.474** [0.359, 0.579], F1@0.25 0.633; R2 tolF1 0.359. GX010137 (0.822) is the corpus's **best** clip; GX010141 (0.510) is above the mean. `[verified]`
- **Opposite failure modes ? the central finding (n=2).** **LOCAL = high recall, low precision; GEMINI = high precision, low recall.** On the normal clip GX010137, local's **recall 0.88 BEATS Gemini's 0.79** (local finds *more* real rallies) but local fires 14 false positives; Gemini fires **zero** false positives but misses 7 rallies. They are **complementary** (on GX010141 they overlap on only ~3 rallies yet each hits ~13?15 of the golden set). ? **fusion (local's recall + Gemini-style precision filtering) is the clear win.**
- **The gaps, re-prioritized by the data:**
  - **G2 ? precision is LOCAL's #1 gap** (over-fires on both clips: 11/22 and 14/39 false positives). This is the cheapest + highest-leverage fix (Gate-A / colocation). Filter local's FPs on GX010137 and it lands ~0.88+ ? at-par with Gemini, since local already out-recalls it.
  - **G1 ? recall** is clip-dependent: a local *strength* on normal play (? Gemini), but on hard continuous-drilling clips *both* miss ~half (GX010141 recall ~45?52%). The recall lever matters mainly for the hard clips.
  - **G3 ? rally-END over-extension** (quick-return-after-point): shuttle keeps moving ? window runs long / merges; R4 (shuttle-only) can't catch it. Contributes to G2's false positives/merges.
- **Local's gap to Gemini SHRINKS on normal play:** hard clip ??0.16 (0.51 vs 0.67) ? normal clip ??0.07 (0.82 vs 0.89), and local *out-recalls* Gemini there. The deficit is precision, not detection.
- **What is NOT the gap:** boundary precision. When local hits a rally its edges are tight (|?start|/|?end| ? ~0.8 s, on GX010137 even tighter than Gemini). The R2/R3/R4 windowing is doing its job. `[verified]`
- **Long-rally lens refinement (P7, n=15, `[verified]`):** filtering the eval to >5s rallies (the eye-catching highlights) shows local's single-court over-fire is **short-FP-dominated** (seg_ratio 1.43?1.04 ? duration-filterable), but **multicourt still over-fires 1.58× on LONG rallies** (adjacent-court games are genuine long rallies). ? the multicourt precision gap needs **court-localization (Gate-B / court mask, Lever C)**, NOT a duration filter; and the strict tolerance precision is gated by **over-merge contamination** of the long-pred set, not just short FPs. See [`QUALITY_ITERATIONS.md`](QUALITY_ITERATIONS.md) "P7".
- ? **Data caveat:** GX010141 was labeled before ~3 rally-annotator-tool bugs were found + fixed (GX010137 is post-fix). GX010141's numbers may shift on a re-export ? re-verify before drawing hard conclusions from it.

4. Design / architecture ? the levers (REUSE MAP)
The signal mechanics live in [`MULTI_SIGNAL_FUSION_PLAN.md`](MULTI_SIGNAL_FUSION_PLAN.md); this is which lever attacks which gap.

- **Lever A ? player-state rally-END rule** (targets **G3 + G2**) `[design]`. Generalize

R4 from shuttle-only to **shuttle OR bilateral player motion**: end the rally at the last frame with competitive evidence from *either*, and **hard-terminate on a "shuttle picked up" event**. Two cheap detectors, no new model required:
55 - *Player kinematics* ? person velocity drops / players go stationary after a point (the casual returner taps it while the other walks). **Reuse:** the person cue producer (torchvision + ByteTrack) in `yolo_hybrid` (fusion P1, `[verified]` exists).
56 - *Shuttle-in-hand / out-of-play* ? `shuttle_player_colocation` (shuttle box ? person box = held) `[verified]` built in `fusion_features.py` (default-OFF, unmeasured on golden) from the fusion plan §4, **or** the no-new-model proxy: **sustained WASB `Visibility=0`** after the last fast stroke (held shuttle goes dark). Use it as a **termination anchor + reverse-timeline search** to the true end (last net-crossing / last fast-shuttle / last bilateral motion) ? the owner's idea, refined: the hold event is *late*, so anchor-then-backtrack, don't use its timestamp as the end.
57 - **Lever B ? recall** (targets **G1**) `[design]`. Both systems miss real rallies ? detector sensitivity: proposal-windowing / velocity-threshold tuning, and the **complementary-fusion** angle below. Measure recall floor per video; the highest-leverage labels are the *missed* rallies (active learning, RALLY_QUALITY_RESEARCH §4.3).
58 - **Lever C ? signal/ensemble fusion** (targets **G1**) `[design]`. Local & Gemini (and shuttle vs person cues) hit *different* rallies ? a union/ensemble lifts recall. Already-built scaffolding: `rally_gate.py` **Gate A** (net-crossings, structural rally-state, `[verified]` built, eval-only ? wire to prod = cheapest precision win for G2) and the FusionSegmenter feature path.
59 - **Pure-precision cleanup** (**G2**): `rally_gate` Gate A kills single-feed / held-shuttle *starts* structurally (?0?1 net crossings vs a real rally's several) ? the cheapest G2 win and independent of the person stack.
60

9. assistant (2026-06-19T17:18:22.279Z)

Both are clearly live/tracked. Let me read the next docs: RALLY_DETECTION_QUALITY_REPORT, PROMINENT_COURT_DETECTION, and the others.

10. user (2026-06-19T17:18:23.704Z)

1 # Local, Cheap, AI Rally Detection for Racket Sports ? Quality Report (Checkpoint)
2
3 **Internal · honest measured results only · checkpoint 2026-06-17 · supersedes the 2026-06-14 draft**
4 **Project: badminton-highlight-indexer · toward a paper**
5
6 > *An honest empirical study of a shuttle-trajectory rally segmenter, its over-segmentation fix, a
7 > task-faithful evaluation metric, a disciplined small-N promotion gate, and a first real-footage
8 > validation against human ground truth and a cloud VLM.*
9
10 ---
11
12 ## Abstract
13
14 We present a fully local, low-cost rally-detection system for racket sports (badminton first) that
15 segments a match video into rally `[start, end]` intervals. The system is built on a **frozen
16 shuttle-trajectory substrate** (a WASB CNN that emits per-frame shuttle position), from which a
17 velocity signal seeds recall-oriented candidate windows that a tiny pure-numpy logistic-regression
18 scorer grades. Every additional cue enters as a confidence-weighted **feature column** the scorer
19 learns to weight ? never a hand-coded branch ? and ships **default-OFF**, promoted only on a measured
20 held-out lift. We evaluate against an owned golden corpus, now **n=11 labelled matches** (grown from
21 n=6 this cycle, including **5 real owner-hand-labelled ground-truth clips**), treating

the labelling
vendor as an interchangeable oracle: the contract is with the golden labels, not their provenance.

Since the last checkpoint (2026-06-14) the headline result moved from "one promoted-then-reverted cue"
to a **statistically significant over-segmentation fix**. Bounded windowing plus a rally-END
inactivity trim (the "cap=6 + R4" milestone) lifts our **task-faithful tolerance-match F1 from 0.091
to 0.323** on n=11 (? +0.232, 95% CI [+0.153, +0.304], paired sign-test **10/0, p=0.002**), drives the
merge-rate from 0.694 to 0.150 (rallies-swallowed-by-a-merge, down on **all 11** videos), and
tightens the mean rally-END error from **52.1 s to 3.8 s**. A first ground-truth comparison against a
cloud VLM (Gemini) shows local reaching **~86?88 % of Gemini on rally detection** (F1@0.25) with the
rally-**START already at parity** ? the entire residual gap is the rally-END. Negative and null
results are reported in full (person and audio fusion cues remain non-significant at n=6; a
net-crossing gate that wins offline does **not** transfer to the served operating point). We also
introduce a deliberately honest **task-faithful evaluation metric** (asymmetric tolerance-match with
strict 1:1 assignment) after finding that plain temporal-IoU is the wrong ruler for this task in both
directions. The contribution is less any single number than a **method for being honest at small
scale** ? and the path to a publishable result is now a short one: more labelled matches.

Headline results at a glance

Over-segmentation milestone (n=11, R2 tolerance-F1)	**0.091 ? 0.323** · ? +0.232 · CI [+0.153, +0.304] · sign-test **10/0, p=0.002**

11. user (2026-06-19T17:18:24.023Z)

Prominent-Court Detection ? kill background-court false positives (multicourt)

> **Status:** `DRAFT` ? design + experiment **owner-gated for any default flip** · **Owner:** Avi · **Created:** 2026-06-19 · **Last updated:** 2026-06-19
> **Lifecycle:** `DRAFT ? IN PROGRESS ? DONE ? archived` (move to `docs/archives/` when DONE)
> **Tracking anchors:** §8 progress tracker is the source of truth; pointer in memory `current-state`.
> **Relation to existing docs:** attacks gap **G2** ("precision ? local over-fires, worst on multicourt") from [`RALLY_DETECTION_GAP_CLOSURE.md`](RALLY_DETECTION_GAP_CLOSURE.md); it is the **shuttle-side** counterpart of the person-side **Gate B / court mask** in [`MULTI_SIGNAL_FUSION_PLAN.md`](MULTI_SIGNAL_FUSION_PLAN.md). Complements the held-shuttle **`shuttle_player_colocation`** (G3) ? court-localization (G2) and held-shuttle (G3) are independent precision levers.
> **Honesty note:** claims are tagged `[verified]` (checked vs code/measured) / `[design]` (proposed). Every signal lands **flag-gated, default-OFF**, promoted only on measured lift per [`EXPERIMENT_HARNESS.md`](EXPERIMENT_HARNESS.md) + [[decision-rigor-norm]].

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0. TL;DR

In **multicourt** footage the camera sees several courts. The shuttle detector (WASB) tracks *any* shuttle in frame, so a rally on a **background / adjacent court** becomes a predicted rally on the **main match** ? a false positive. The fix: identify the **prominent court** (the foreground court that occupies the majority of the frame, where the match being filmed is played) and **keep only rallies whose shuttle lives inside it**. Mechanically this re-defines the shuttle signal from `(x, y, visible)` to `(x, y, visible-AND-inside-the-prominent-court)` *before* windowing ? so a background-court shuttle reads as "not there" and never forms a rally. A flat 2D floor polygon is the MVP; a **pseudo-3D play-volume** (court floor + the air column above it) is the refinement so a high clear over the prominent court isn't wrongly rejected. Lands as a **default-OFF experiment**; the flip is gated on a Launch Report ([[launch-report-convention]]).

1. Problem & motivating example `[verified]`
- **Measured gap (G2):** local over-fires ~2× vs Gemini, and the over-fire is **worst on multicourt** ? even among long (>7s) rallies, multicourt over-fires **1.58×** (background-court games are *real* long rallies, just on the wrong court). See `RALLY_DETECTION_GAP_CLOSURE.md` §3 + the P7 long-rally lens.
- **Concrete anchor ? GX010142 rally #1 (2026-06-19).** Owner observation: *"the first rally is just a shuttle flying in the non-prominent court? while the prominent-court players have not started."* Confirmed from the cached WASB trajectory (`output/wasb/GX010142_1080p__*_wasb.csv`): rally #1 (`0.72?3.65s`, 2.94s) has a shuttle centroid at **meanX ? 1616** on a 1920-wide frame (?84% to the right) ? spatially apart from the central cluster where most rallies live (meanX ? 1000?1300). Several other suspicious rallies (#3, #9, #12, #16) cluster in the same far-right band ? consistent with a **right-side adjacent court**.
- **Why a 1D threshold is not enough `[verified]`:** the courts do not separate cleanly on X or Y alone (rally Y-centroids overlap 250?620 px across courts; the far-right band mixes with foreground play). The courts are **2D regions** under perspective ? so we need the court *geometry*, not a coordinate cutoff. This is the owner's "pseudo-3D / court model" intuition, validated by the data.

2. Goal & non-goals
- **Goal:** on multicourt clips, drop rallies whose shuttle is outside the prominent court, **without losing real prominent-court rallies** ? measured per-video, no regression on single-court clips (where it must be a no-op or a do-no-harm).
- **Non-goals (v1):** multi-camera; tracking players across courts; full court-line homography for scoring; anything biometric. Single-court footage is explicitly a **no-op** (one court ? everything is "prominent").

3. Decisions locked
#	Decision	Rationale
D1	The prominent court is a **2D region (polygon) in image space**, refined to a **pseudo-3D play volume**	1D thresholds don't separate courts under perspective (§1)
D2	Gate the **shuttle trajectory** by the region **before windowing** (re-define visibility)	Stops a background rally at the source ? no window ever forms there. Reuses `point_in_polygon`
D3	**Default-OFF, flag-gated; single-court = no-op**	Zero-regression rollout ([[weak-signals-retained]]); no behaviour change unless a polygon is supplied + the flag set
D4	Reuse the existing **`court_polygon`** config + `point_in_polygon`; do NOT invent a new geometry stack	`FusionConfig.court_polygon` + `fusion_features.point_in_polygon` already exist `[verified]`
D5	The prominent court can be **supplied (config) OR auto-derived**; auto-derivation is its own measured step	Config polygon is the cheapest first experiment; auto is the product path

4. Design

4.1 The signal change (the core experiment)
Today the windower consumes the raw WASB trajectory: a per-frame `(Frame, Visibility, X, Y)`. The change is a **pre-windowing filter**:

```
for each frame f:  if shuttle_visible(f) and NOT inside_prominent_court(X_f, Y_f):  mark f as NOT visible
```

```

A background-court shuttle is then indistinguishable from "no shuttle" ? the velocity windower never opens a rally there. This is purely **subtractive** on the shuttle stream (it can only remove visibility, never add it), so on a correctly-drawn prominent court it cannot manufacture rallies ? only suppress out-of-court ones. (Mirror of `rally_gate`'s "do-no-harm" discipline.)

### 4.2 Identifying the prominent court (four routes, cheapest first)
1. **Configured polygon** (MVP, `[design]?buildable now`): a normalized `[[x,y],?]` court polygon per camera/clip, exactly the existing `FusionConfig.court_polygon`. Draw it once from the dense shuttle/player cluster. Good enough to validate the whole concept on GX010142 + the multicourt golden clips.
2. **Player-location auto-derivation** (`[design]`): run `PersonDetector.detections_per_frame` `[verified] exists`, take the **largest, most-central, most-persistent** player cluster (the foreground players are bigger boxes lower in frame), and fit the prominent-court polygon to where *those* players' feet live. "Prominent = where the match's players are." This is why the owner asked to **extract person features** ? players anchor the court.
3. **Shuttle-density auto-derivation** (`[design]`): the prominent court is the densest sustained shuttle-activity region; cluster the visible trajectory and take the dominant mode. Person-free fallback.
4. **Court-line detection** (`[design]`, heaviest): Hough/segmentation court lines ? the largest court quad. Most accurate, most work; deferred.

v1 ships route 1 (config) + measures; route 2 (player-anchored) is the product upgrade.

### 4.3 Pseudo-3D play volume (the refinement)
A flat floor polygon rejects a shuttle at the top of a high clear *over the prominent court* (it projects above the floor quad). The pseudo-3D model treats the prominent court as a **floor quad + a vertical air column**: accept a shuttle if its image point lies within the floor polygon **expanded upward** by a height envelope (the max plausible shuttle height projected through the camera). Practically: the floor quad plus an upward `pad` (perspective-aware ? larger near the net/far baseline). Start with a generous fixed upward pad (cheap, MVP); upgrade to a per-camera height envelope only if measurement shows real rallies being clipped. This keeps the floor tight (rejects adjacent-court floor play) while not clipping legitimate high shuttles.

### 4.4 Reuse map `[verified]`
| Need | Existing code |

#### 12. assistant (2026-06-19T17:18:27.557Z)

Now let me read the progress trackers / status sections of the remaining docs and the report's status footer. Let me check QUALITY_ITERATIONS, R2_EVAL_METRIC_DESIGN, MULTI_SIGNAL_FUSION_PLAN headers, and the progress trackers (§7/§8) of the two tracked docs and the report's conclusion.

#### 13. user (2026-06-19T17:18:28.517Z)

# Rally-detection quality iterations ? living log

**Purpose.** A running, reverse-chronological log of how we evolve rally-detection quality: each
iteration's hypothesis, what shipped, the measured (or expected) effect, and the lesson. This is the
**live** continuation of the historical trail in
[`archives/quality-iterations/`](archives/quality-iterations/README.md) ? per the archive policy
(`CLAUDE.md`), **terminal-state** (DONE/REJECTED) work gets a dated subdir there; this file stays at
`docs/` top level and keeps growing, then its entries graduate to the archive when they conclude.

**Methodology (non-negotiable).** Every quality change rides the experiment harness
([EXPERIMENT_HARNESS.md](EXPERIMENT_HARNESS.md)): new cues land **flag-gated, default
```

OFF**; they're
graded **offline against golden labels** (per-video + LOVO-honest); promotion to default happens **only
on measured lift** through the `run_regression.py` gate; **rollback = flip a config flag, never
`git revert`.** Cues are designed in MULTI_SIGNAL_FUSION_PLAN.md; scoring
in EVALUATION.md; the 2-layer model in
HOW_RALLY_DETECTION_WORKS.md.

Golden corpus = the 15 labelled matches in `output/human_lovo_manifest.json` (the original 6 ?
`adarsh_avi_singles, kushagra_singles, largetest_doubles, gbaaddy, testlarge_short,
mahadevpura_singles` ? owner-confirmed 2026-06-13, since grown to 15 with the later GoPro/Boxhill clips). The model trained on these is what we run alongside the existing
detector on NEW footage for the side-by-side review.

Baselines that drive strategy (IoU 0.5 vs human golden labels)

Path	Mahadevpura (clean-ish)	Kushagra (multi-court)
Heuristic (velocity windowing)	0.657	0.157
Gen-0 owned head (LOVO, n=6)	0.744	0.561
Two-stage WASB+Gemini refine	0.83	?
Gemini wrapper (`gemini-flash-latest`)	0.93	0.66

Reading: clean footage is commoditized (serve via Gemini); **multi-court drops the commodity wrapper
to 0.66 ? that is the owned model's ship target.** Baselines tracked in `eval_baselines/`. The owner's

14. user (2026-06-19T17:18:29.453Z)

R2 ? Task-faithful rally-boundary eval metric (tolerance-match + over-seg guard)

Status: **IMPLEMENTED (augmenting), 2026-06-17** ? owner-reviewed + decisions locked 2026-06-17;
`backend/eval/tolerance_metrics.py` + `rally_seg_eval` reporting landed (default-additive; tIoU stays the
gate pending the ablation experiment). Gates the rally-end work (R4) and the dual-eval-set regression gate.
A focused increment of the rally-quality project (RALLY_QUALITY_RESEARCH.md §7;
the day-1 validation that motivates it: REAL_FOOTAGE_VALIDATION_REPORT.md;
the directive: `eval-boundary-tolerance` + `eval-dual-set-regression`).

> **Implementation refinement (2026-06-17):** §2.3.2's boundary-error distribution is computed over
> **unconditional best-overlap** matches, NOT the within-? 1:1 matches ? a ?-gated set is survivorship-biased
> (loose ends fail the gate and vanish, hiding the very end-deficit the diagnostic exists to surface; that
> within-? deficit instead shows up as low `tol_recall`). Matches the day-1 validation method. Measured on the
> n=9 corpus: **|?start| 1.25 s vs |?end| 2.19 s** ? "start?solved, end=the gap", confirmed.

---

0. TL;DR
`tIoU-F1@0.5` is the wrong ruler for rally detection, in **both** directions: it over-penalizes the

intrinsically fuzzy rally START (the ~1?1.5 s service window), and its overlap matching is murky about
over-production. R2 replaces the headline with a **tolerance-match metric under strict 1:1 (Hungarian)
assignment**, reported as a **decomposition** ? `P@? / R@? / F1@?` + **boundary-error distributions split
into start/end** + an **over-segmentation guard** + a **?-sweep** ? not a single number.

Honest, slightly uncomfortable consequence (measured, see §3): under R2 the local detector's headline
number *drops* and the gap to Gemini *widens* vs what tIoU showed ? because R2 faithfully charges for the
1.5× over-production and the loose ends that tIoU's overlap matching partly absorbed. That is the point:
R2 splits "behind Gemini" into its two real causes ? **(a) over-production [precision]** and **(b) loose
rally-ends [the R4 lever]** ? and forgives only the inherent start fuzziness. It is a more honest ruler,
not a flattering one.

---

1. Why `tIoU@0.5` is the wrong ruler (grounded in the n=2 ground-truth clips)
1. **Over-penalizes the START.** For a 5.7 s rally, `tIoU ? 0.5` tolerates only ~±1.9 s of boundary shift
   (`? = D·(1??)/(1+?)`). The service window (stance ? racquet contact) alone is ~1?1.5 s ? even **Gemini's**
   median |?start| is 0.98?1.46 s. So tIoU spends most of its budget on a boundary that is *intrinsically*
   undefined to ~1 s. (Owner directive: don't strictly align starts; tolerate ~±2 s.)
2. **Count parity is a deceptive proxy.** mahadevpura-2 hit 1.03× of Gemini's count yet F1@0.5 = 0.150 ?
   right count, wrong boundaries. A count- or overlap-based view hides this.
3. **It flips config rankings on noise.** The hard-cap was golden-neutral at n=6 (+0.006 n.s.) but "best" at
   n=7 once a continuously-active clip entered ? a metric artifact at the 0.5 cliff, not a quality change.
4. **The deficits we must see are start?end-symmetric.** Local's |?start| ? Gemini's; the **entire** gap is
   the rally-**END** (|?end| 2.1?3.1 s vs Gemini ~0.8?1.2 s). A single blended IoU cannot show that ? and
   directing R4 needs exactly that decomposition.

15. user (2026-06-19T17:18:30.141Z)

Multi-signal rally fusion ? add a human (person) cue alongside the shuttle

Status: **P1 DONE** (2026-06-13) ? the person cue is extracted to
`backend/pipeline/detectors/person_detector.py` (`PersonDetector`), consumed by `yolo_hybrid` with no
behavior change. **P2?P4 remain owner-gated.** Phases below are sequenced but each is a separate PR.
Owner decisions captured in [§9](#9-open-decisions-for-the-owner).

> **Companion docs:** the 2-layer mental model is HOW_RALLY_DETECTION_WORKS.md;
> the modular-fusion architecture this builds on is **ADR-007** (archives/decisions/DECISIONS.md);
> the permissive-dependency rule is **ADR-002 / ADR-005**; the audio cue (the #1 next cue, already
> scaffolded) is archives/research/AUDIO_RALLY_DETECTION_PLAN.md.
> The vendor labels that gate the learned path come from GOLDEN_SET_VENDORING.md
> and the labeling spec in ROORA_LABELING_GUIDELINES.md.

```
> **How each cue here is added/tested without production regression** ? A/B, shadow eval, and the
> promote-only-on-lift discipline ? is [EXPERIMENT_HARNESS.md](EXPERIMENT_HARNESS.md) (every P1?P4 cue
> below lands flag-gated, default OFF, promoted only on measured LOVO lift).

---

## 1. Context ? why this, why now

Rally detection today runs **one cue at a time**. Each segmenter independently turns a single signal
into rally windows:

| Segmenter | Signal | File |
|---|---|---|
| `wasb_hybrid` / `tracknet_hybrid` | shuttle velocity (frozen WASB/TrackNet trajectory) | `backend/pipeline/detectors/tracknet_runner.py` |
| `yolo_hybrid` (the Pydantic `default_segmenter` fallback; `config.json` overrides it to `gemini`) | person velocity (torchvision + ByteTrack) | `backend/pipeline/segmenters/yolo_hybrid.py` |
| `motion` | raw pixel motion | `backend/pipeline/segmenters/motion.py` |
| `gemini` | a cloud VLM watching the whole clip | `backend/providers/gemini.py` |

**There is no fusion.** The owner's directive: make the detector **use the shuttle *and* the humans
together** ? and be ready to fold in more signals (audio, later pose) ? delivered
**MIT / Apache-2.0 / BSD-clean** so we can ramp commercially without a licensing rethink.

This is **not greenfield** ? it operationalizes work the repo already blessed:

- **`HOW_RALLY_DETECTION_WORKS.md` open question:** *"Fusion: combine shuttle-motion (WASB) with
  player-motion for static cams."*
- **ADR-007** already designed a **modular "rally layer"** that fuses cues *outside* the frozen WASB
  model, with **audio as the #1 next cue and person/pose parked as #2**. This plan picks up the parked
  person cue and generalizes the fusion seam so future cues are "register a producer," not "re-touch
  the segmenter."
- **`backend/pipeline/detectors/rally_gate.py` already exists** ? the net-crossing gate, explicitly
  labelled in its own docstring as *"Gate A in the over-fire fix spectrum (**B = player-stance state
```

16. assistant (2026-06-19T17:18:34.088Z)

Now let me check the RALLY_DETECTION_QUALITY_REPORT's status/conclusion footer and the progress trackers (§7 of RALLY_QUALITY_RESEARCH, §7 of GAP_CLOSURE, §8 of PROMINENT_COURT). Also need to grep for inbound references and check archives for existing copies.

17. user (2026-06-19T17:18:36.657Z)

```
quality-iterations
AUDIO_RALLY_DETECTION_PLAN.md
RALLY_DETECTION_RESEARCH.md
```

18. user (2026-06-19T17:18:36.746Z)

```
docs\archives\COMMERCIAL_READINESS_REVIEW.md
docs\archives\MONETIZATION_AUDIT.md
docs\archives\checkpoints\2026-06-eval-baseline\README.md
```

```
docs\archives\checkpoints\2026-06-logging-reporting\README.md
docs\archives\checkpoints\2026-06-scaffolding-complete\README.md
docs\archives\past_projects\README.md
docs\archives\research\AUDIO_E1_FINDINGS.md
docs\archives\research\AUDIO_RALLY_DETECTION_PLAN.md
docs\archives\research\RALLY_DETECTION_RESEARCH.md
docs\archives\checkpoints\2026-06-strategy-foundation\README.md
docs\archives\quality-iterations\2026-06-human-gt-tuning\README.md
docs\archives\quality-iterations\2026-06-initial\README.md
docs\archives\quality-iterations\2026-06-multivideo-lovo\README.md
docs\archives\research\OWNED_MODEL_TRAINING_STUDY.md
docs\archives\past_projects\low-regret-moves\LOW_REGRET_MOVES_PLAN.md
docs\archives\decisions\DECISIONS.md
docs\archives\TEST_RUN_SUMMARY.md
docs\archives\quality-iterations\2026-06-first-ab-experiment\README.md
docs\archives\checkpoints\2026-06-14-hardening-loop-complete.md
docs\archives\CODE_AUDIT_AND_TEST_HARDENING-COMPLETED-2026-06-14.md
docs\archives\DEFERRED_HARDENING_PLAN-COMPLETED-2026-06-14.md
docs\archives\HARDENING_LOOP_HANDOFF-COMPLETED-2026-06-14.md
docs\archives\README.md
docs\archives\quality-iterations\2026-06-18-lovo-resweep-n13\README.md
docs\archives\quality-iterations\README.md
```

19. assistant (2026-06-19T17:18:39.962Z)

None of my 7 batch docs have an already-archived copy with the same stem (the archived `RALLY_DETECTION_RESEARCH.md` is a different doc from `RALLY_DETECTION_GAP_CLOSURE.md`/`RALLY_QUALITY_RESEARCH.md`). Now let me grep for inbound references to each filename across the repo, and check the status footers of the report + progress trackers.

20. user (2026-06-19T17:18:42.019Z)

```
CLAUDE.md:2
docs\ANALYZERS_AND_RECONCILERS.md:6
docs\CODE_MAP.md:1
docs\DESKTOP_APP_PLAN.md:1
backend\config\models.py:3
docs\GOLDEN_DATA_SHARING.md:1
eval_baselines\fixtures\README.md:1
docs\EXPERIMENT_HARNESS.md:4
artifacts\pr245_review.md:2
docs\HOW_RALLY_DETECTION_WORKS.md:1
docs\archives\CODE_AUDIT_AND_TEST_HARDENING-COMPLETED-2026-06-14.md:1
artifacts\pr245_review.json:2
artifacts\pr245_fixes_plan.md:1
docs\GOLDEN_VIDEOS.md:1
backend\pipeline\detectors\person_detector.py:1
backend\pipeline\detectors\court_gate.py:1
docs\GOLDEN_REGRESSION_FIXTURES.md:3
backend\pipeline\detectors\tracknet_runner.py:1
docs\PERSONALIZATION_PLAN.md:1
docs\NEXT_STEPS.md:3
docs\RALLY_DETECTION_GAP_CLOSURE.md:7
docs\R2_EVAL_METRIC_DESIGN.md:4
docs\VIDEO_RECORDING_GUIDELINES.md:2
docs\RALLY_QUALITY_RESEARCH.md:12
docs\REAL_FOOTAGE_VALIDATION_REPORT.md:2
docs\R1_MILESTONE_LAUNCH_REPORT.md:1
docs\README.md:7
backend\eval\windowing.py:1
docs\ROORA_LABELING_GUIDELINES.md:2
docs\QUALITY_ITERATIONS.md:5
backend\eval\tolerance_metrics.py:1
docs\PROMINENT_COURT_DETECTION.md:4
```

```
tests\test_config.py:1
docs\PROJECT_DOC_TEMPLATE.md:1
docs\POST_166_WORK_ITEMS_MILESTONES_AND_RISK_ASSESSMENT.md:1
tests\test_court_gate.py:1
docs\archives\quality-iterations\2026-06-18-lovo-resweep-n13\README.md:3
backend\eval\rally_seg_eval.py:3
scripts\nightly_regression.py:4
backend\eval\experiment.py:1
backend\eval\golden_fixtures.py:1
docs\archives\quality-iterations\2026-06-first-ab-experiment\README.md:1

Found 102 total occurrences across 42 files.
```

21. assistant (2026-06-19T17:18:45.412Z)

```
High inbound reference counts across code and docs. These are deeply wired-in live docs. Let me
confirm the per-filename "cited as live" status by checking NEXT_STEPS and CLAUDE.md
specifically, and verify the report's status footer plus the progress trackers to confirm none
claim terminal state.
```

22. user (2026-06-19T17:18:47.489Z)

```
17:  `docs/HOW_RALLY_DETECTION_WORKS.md` (2-layer mental model) ?
`docs/MULTI_SIGNAL_FUSION_PLAN.md`
60:- **Substantial design/project work gets a tracked project doc** following
`docs/PROJECT_DOC_TEMPLATE.md` (status header · decisions-locked · progress tracker ·
definition-of-done; archived on completion). Precedents: `docs/RALLY_QUALITY_RESEARCH.md` §7,
`docs/GOLDEN_REGRESSION_FIXTURES.md`.
```

23. user (2026-06-19T17:18:49.174Z)

```
59:- **Docs:** `docs/MULTI_SIGNAL_FUSION_PLAN.md` (shuttle + person + audio, one fusion layer) ·
70:- **External due-diligence review inputs (2026-06-17) ? PICK UP AFTER current ideas are
exhausted; full details + citations in [`RALLY_QUALITY_RESEARCH.md`](RALLY_QUALITY_RESEARCH.md)
§5.6.** An owner-commissioned 31-reference review of the quality report *validated* the
methodology and surfaced these **future** levers (research inputs, not active work, not new
claims; verify citations before any submission):
96:   (`RALLY_QUALITY_RESEARCH.md` §5.6) ? better fit for the cheap/local/fast mandate.
```

24. assistant (2026-06-19T17:18:52.211Z)

```
Both CLAUDE.md and NEXT_STEPS.md cite these as live/authoritative. Now let me verify the
report's conclusion/status (does it claim COMPLETED?) and the progress trackers of the tracked
docs to confirm none are terminal.
```

25. user (2026-06-19T17:18:55.542Z)

```
=== QUALITY_REPORT tail ===
- **Lead with the tolerance-match metric + over-segmentation metrics** (merge-rate, F1@k), not
bare tIoU.
- **Per-cue ablation with CIs** on the grown corpus (the `ablation.py` harness already exists).

**Target venues** (applications/systems-oriented): **CVIU, ICIP, IEEE Trans. Multimedia, or a
CVPR/
ICCV workshop** (CV-for-sports / fine-grained video). Not the CVPR/ICCV main track (no
algorithmic
novelty). The cheapest path to a publishable result remains **more labelled matches**, not more
engineering.

---

10. Engineering & reproducibility

- **Experiment leaderboard** ? every `compare()` appends one reproducible row to
```

`eval_baselines/experiment_leaderboard.json`, pinned to `(spec_hash, label_set_hash, IoU)`, so any
  regression traces to a recipe change, not drift.
- **Nightly regression tripwire** ? CPU-only (cached trajectories), guards DEFAULT (served) and MILESTONE
  configs; exit 1 = regression, 4 = stale baseline.
- **Standing litmus** ? `served_gate_a.py` pins the honest eval?serve facts.
- **CI gates** ? Ruff, lenient-but-green mypy with a ratchet, syntax+import smoke (no GPU stack), and the
  full offline mocked suite with a coverage floor. Combined with **default-OFF cues + a pinned CONTROL
  variant**, rollback is a config flip, never a `git revert`.
- **Privacy-safe fixtures** ? golden-regression fixtures + de-attribution tooling allow video-free
  shipping of the eval sets.

---

11. Conclusion

The contribution of this work is less any single F1 number and more a **method for being honest at small
scale** ? and this cycle shows that discipline paying off. On an owned golden corpus grown to **n=11**
(now with real ground truth), bounding the windowing and trimming the rally-END produced the project's
first statistically significant rally-quality win (tolerance-F1 0.091 ? 0.323, p=0.002), drove the
merge-rate down on **every** video (0.694 ? 0.150), and ? measured against human ground truth for the
first time ? put the rally-**START at parity with a cloud VLM**, localizing the **entire remaining gap to
the rally-END**. We reported the nulls in full (person, audio), banked the eval?serve lesson (the
reverted Gate-A ? the honest W2), and introduced the task-faithful metric that made the over-production
tIoU hid finally visible. The architecture stays deliberately standard where the literature is mature; we
claim novelty only on the task-faithful metric and the small-N promotion discipline. The path to a
publishable result is now short and concrete: **more labelled matches, multi-court isolated, ablations on
the grown corpus** ? the moat was always the consented data plus the honest harness, not the weights.

=== GAP_CLOSURE §7 tracker ===
4:> **Lifecycle:** `DRAFT ? IN PROGRESS ? DONE ? archived` (move to `docs/archives/` when DONE)
5:> **Tracking anchors:** §7 progress tracker is the source of truth; pointer in memory
`current-state`; add to `docs/README.md` + `docs/NEXT_STEPS.md` when work starts.
65:- **Nothing here is shipped or even started** ? implementation is owner-gated. A DONE row in
§7 means *measured on golden*, not *flipped on by default*.
69:## 7. Deliverables & progress tracker ? **source of truth**
70:Legend: ? Todo · ? In progress · ? Done · ? Gated. One small PR per row, off `origin/master`,
no stacking.
74:| P0 | Golden scoring loop (`ingest_golden.py` + `boxhill17_score_golden.py`); GX010141
ingested (#12) + scored | ? | No | ? | this work |
75:| P1 | Ingest the rest of the Boxhill golden batch (137/020140 + a large-count clip) ? per-
video baseline table | owner labels | ? owner | ? | ? |
76:| P2 | Wire `rally_gate` **Gate A** (net-crossings) to the served path ? cheapest G2 win | ?
| ? owner | ? | ? |
77:| P3 | Extract person-velocity + `shuttle_player_colocation` on the golden clips (default-OFF
features) | P1 | ? owner | ? | ? |
78:| P4 | **Lever A**: player-state rally-END rule (R-next) ? measure ?F1 / ?merge-rate /
?\|?end\| vs R4 | P3 | ? owner | ? | ? |

79:| P5 | **Lever B/C**: recall + complementary fusion ? measure recall lift, no precision regression | P3 | ? owner | ? | ? |
80:| P6 | Launch decision: milestone + best levers default-ON, with a Launch Report (both rulers + over-seg guard) | P4,P5 | ? owner | ? | ? |
81:| P7 | **Long-rally (>5s) quality lens** (eval `--min-duration`/`--stratify`) + reel knob (`stitching.min_rally_duration`), both default-OFF. **Measured n=15:** single-court over-fire is short-FP-dominated (seg_ratio 1.43?1.04) but **multicourt over-fires 1.58× even on long rallies** (? court-localization, Lever C) and strict tolP doesn't rise (over-merge-contaminated). See [`QUALITY_ITERATIONS.md`](QUALITY_ITERATIONS.md) "P7". | ? | No (eval+cfg, default-OFF) | ? measured | this PR |
88:## 9. Definition of done
93:- [ ] ? flip `Status ? DONE`, archive to `docs/archives/`.

26. assistant (2026-06-19T17:19:03.502Z)

This confirms everything. The GAP_CLOSURE doc explicitly states "Nothing here is shipped or even started ? implementation is owner-gated" and its §7 tracker has mostly ?/? rows ? clearly LIVE in-progress work. The QUALITY_REPORT is a dated checkpoint with future-facing next steps (no COMPLETED/terminal header). All 7 docs are LIVE.

Let me do a final confirmation on the report's status header and whether any doc is a duplicate of an archived copy ? none have matching archived stems. The report header says "checkpoint 2026-06-17 · supersedes the 2026-06-14 draft" ? it supersedes an earlier draft but is itself the current checkpoint, not superseded. All evidence gathered.

27. user (2026-06-19T17:19:40.187Z)

Structured output provided successfully

28. assistant (2026-06-19T17:19:51.311Z)

All 7 docs in my batch are KEEP_LIVE. Summary of findings:

- **RALLY_QUALITY_RESEARCH.md** ? ? Foundation/tracked-project doc, definition-of-done not yet met, cited as authoritative in CLAUDE.md + NEXT_STEPS.md (12 inbound refs). LIVE.
- **RALLY_DETECTION_GAP_CLOSURE.md** ? DRAFT execution tracker; §6 explicitly says "Nothing here is shipped or even started," §7 rows mostly ?/? owner-gated. LIVE in-progress.
- **RALLY_DETECTION_QUALITY_REPORT.md** ? current 2026-06-17 paper checkpoint that *supersedes* the older draft (it is not itself superseded); forward-looking §9/§11 + active paper aim. LIVE.
- **PROMINENT_COURT_DETECTION.md** ? brand-new (2026-06-19) DRAFT design/experiment, owner-gated, actively being built (`court_gate.py` exists). LIVE.
- **QUALITY_ITERATIONS.md** ? self-described living log that "stays at docs/ top level and keeps growing"; only concluded entries graduate to dated archive subdirs. LIVE by design.
- **R2_EVAL_METRIC_DESIGN.md** ? "IMPLEMENTED (augmenting)" but not terminal ("tIoU stays the gate pending the ablation experiment"; gates open R4 work); backs live `tolerance_metrics.py`. LIVE.
- **MULTI_SIGNAL_FUSION_PLAN.md** ? only P1 DONE, P2?P4 owner-gated; cited live in CLAUDE.md + NEXT_STEPS.md. LIVE.

No already-archived duplicates exist for any of these stems (the archived `RALLY_DETECTION_RESEARCH.md` is a distinct doc). No DEDUPE_STUB or ARCHIVE_NOW cases, and no contradiction traps requiring ASK_OWNER ? none of the 7 claim a terminal COMPLETED/DONE/REJECTED state while being cited live. Nothing to archive from this batch.